

From Friction to Architecture

Problem Framing,
Needs Assessment, & Requirements Elicitation

DAEN 459 - Fall 2026
Alexander Abuabara



The Discovery Workflow

- **Problem Framing:** Defining the right challenge.
- **Needs Assessment:** Identifying the foundational gaps.
- **Requirements Elicitation:** Gathering specific technical rules.

1. Problem Framing

Concept: Shifting from surface-level symptoms to root business challenges. Avoids building expensive pipelines for the wrong issue.

Example:

- *Symptom:*
"The dashboard is too slow."
- *Poor Framing:*
"We need to rewrite the entire ETL pipeline in Spark."
- *Good Framing:*
"Business analysts lose decision-making time because downstream BI tables lack optimized partitioning and indexing for morning reports."

2. Needs Assessment

Concept: Evaluating the current state vs. the desired future state. Separates user “wants” from systemic “needs.”

Example:

- *User Want:*
“We want real-time streaming data for all sales metrics.”
- *Assessment Finding:*
The finance team only reviews reports weekly. The underlying source systems update via daily batch jobs.
- *The True Need:*
A robust, automated daily batch pipeline with data quality checks, rather than an expensive Kafka infrastructure.

3. Requirements Elicitation

Concept: Actively extracting detailed, actionable constraints (functional and non-functional) from stakeholders and systems.

Example:

- **Functional:**

The pipeline must ingest JSON files from an S3 bucket, schema-validate them, and flatten them into a Snowflake warehouse.

- **Non-Functional (SLA):**

Data must be available by 6:00 AM UTC daily; the pipeline must handle up to 500% traffic spikes during Black Friday.

Summary Checklist for Teams

1. **Don't write code** during **Problem Framing**.
Focus entirely on the *business* bottleneck.
2. Audit your sources (thoroughly inspecting the data systems you depend on) and destinations during **Needs Assessment**.
3. Lock down metrics, volume, and latency during **Requirements Elicitation**.